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SUBJECT CODE NO: - RNP-8010**FACULTY OF SCIENCE AND TECHNOLOGY****S.E. (Civil Engg.) (NEP) Sem - III****Examination November / December 2025 (To be held in February-2026)****Surveying****[Time: 2:00 Hours]****[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

1. **Part A: Question No 1 is compulsory.**
2. **Attempt any four questions from remaining Part B: Q.2 to Q.7.**

Part A**Q1 Choose the correct alternative for the following questions:****10**

- 1) Levelling is used to determine:

(a) Horizontal distance	(b) Vertical distance
(c) Angle between lines	(d) None
- 2) An instrument used for levelling is:

(a) Theodolite	(b) Dumpy Level	(c) Compass	(d) Clinometer
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- 3) A temporary benchmark is:

(a) Fixed permanently	(b) Established temporarily
(c) Used for all surveys	(d) None
- 4) Contour lines of different elevations never:

(a) Cross each other	(b) Merge
(c) Intersect at water bodies	(d) Are parallel
- 5) Reciprocal levelling is used when:

(a) Instrument is in good condition	(b) Ground is uneven
(c) Long sights are taken	(d) For slope measurement
- 6) Auto level is a type of:

(a) Digital level	(b) Self-levelling instrument
(c) Theodolite	(d) Compass
- 7) Levelling staff is graduated in:

(a) Feet	(b) Inches	(c) Metres	(d) Yards
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- 8) Back sight is the reading taken on:

(a) Benchmark	(b) Turning point
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- (c) Fore station (d) Intermediate point
- 9) Contour lines close together indicate:
 (a) Flat slope (b) Steep slope
 (c) Uniform slope (d) Depression
- 10) The process of checking staff readings is known as:
 (a) Booking (b) Fly levelling
 (c) Check levelling (d) Reciprocal levelling

Part B

Solve any FOUR of the following

20

- Q2** Explain with neat sketch the types of levelling. **05**
- Q3** Describe linear and angular methods of setting out simple circular curves. **05**
- Q4** Write the advantages and disadvantages of Plane table Survey. **05**
- Q5** Explain the characteristics and uses of contour lines. **05**
- Q6** The following staff readings were observed successively with a level the instrument is moved by third sixth and eighth readings. 2.228 :1.606 :0.988 :2.090 :2.864 :1.262 :0.602 :1.982 :1.044 :2.684 m enter the reading in record book and calculate R.L. if the first reading was taken at a B.M of 432.383m **05**
- Q7** Describe the testing and permanent adjustments of Dumpy Level. **05**

